Chapter 2. Multivariate Random Variables

Joint and marginal distributions, moments, conditional expectation, independence, random vectors

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2.1 Joint and marginal distributions

A pair (X, Y) is described by a **joint pmf** $f(x, y) = P(X = x, Y = y)$ in the discrete case, or a **joint pdf** $f \geq 0$ with $P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$ in the continuous case. In both cases $f \geq 0$ and f sums or integrates to 1.

Definition (Marginal distributions)

$f_1(x) = \sum_y f(x, y)$ or $\int f(x, y)\, dy$, and $f_2(y) = \sum_x f(x, y)$ or $\int f(x, y)\, dx$.

Marginals are obtained by "summing out" the other variable; in a table they are the row and column totals. The joint distribution determines the marginals, but not conversely.

The **joint cdf** is $F(x, y) = P(X \leq x, Y \leq y)$. For continuous (X, Y), $f = \partial^2 F / \partial x \partial y$ at continuity points. The marginal cdfs are limits: $F_1(x) = \lim_{y \to \infty} F(x, y)$.

Theorem 2.1

(a) F is nondecreasing in each argument, and for $x_1 < x_2$, $y_1 < y_2$,

$$P(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1) \geq 0.$$

(b) $F(x, y) \to 0$ as $x \to -\infty$ or $y \to -\infty$, and $F \to 1$ as both $\to +\infty$.
(c) F is right-continuous in each argument.

Proof. The rectangle identity is inclusion–exclusion for $\{X \leq x_2, Y \leq y_2\}$ minus the two strips. (b), (c) follow from continuity of probability (Theorem 1.2) as in one dimension. ∎

Example 2.1

Let $f(x, y) = c(x + y)$ on $0 \leq x, y \leq 1$. Then $\int_0^1 \int_0^1 (x + y)\, dy\, dx = 1$, so $c = 1$. The marginal is $f_1(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}$ on $[0, 1]$, and by symmetry $f_2(y) = y + \frac{1}{2}$. Also $P(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y)\, dy\, dx = \int_0^1 \frac{1 - x^2}{2}\, dx = \frac{1}{3}$.

Example 2.2 (A discrete joint distribution)

An urn has 2 red, 3 blue and 4 green balls; draw 3 without replacement, and let X, Y be the numbers of red and blue balls. Then

$$f(x, y) = \binom{2}{x}\binom{3}{y}\binom{4}{3 - x - y} \Big/ \binom{9}{3}, \qquad x + y \leq 3,$$

e.g. $f(0, 0) = 4/84$ and $f(1, 1) = 24/84$. Summing out y (Vandermonde) gives the marginal $f_1(x) = \binom{2}{x}\binom{7}{3-x}/84$, i.e. $35, 42, 7$ over 84 for $x = 0, 1, 2$: the hypergeometric $H(3; 9, 2)$, as it must be.

Example 2.3 (Joint cdf)

For $f(x, y) = e^{-y}$ on $0 < x < y$: $F(x, y) = \int_0^{x \wedge y} \int_s^y e^{-u}\, du\, ds = \int_0^{x \wedge y} (e^{-s} - e^{-y})\, ds = 1 - e^{-x \wedge y} - (x \wedge y) e^{-y}$ for $x, y \geq 0$. Letting $y \to \infty$ gives $F_1(x) = 1 - e^{-x}$, and letting $x \to \infty$ gives $F_2(y) = 1 - e^{-y} - y e^{-y}$, the Gamma$(2, 1)$ cdf.

2.2 Expectations, covariance and correlation

$E[g(X,Y)] = \sum\sum g(x,y)f(x,y)$ or $\iint gf$; it is linear and monotone. If g depends only on x, $Eg(X)$ can be computed from the joint or from the marginal: integrating out y first gives $\int g(x)f_1(x)\,dx$.

Definition (Covariance and correlation)

$\mathrm{Cov}(X,Y) = E[(X-\mu_1)(Y-\mu_2)]$ and, for positive variances, $\rho = \mathrm{Corr}(X,Y) = \mathrm{Cov}(X,Y)/(\sigma_1\sigma_2)$.

Theorem 2.2

(a) $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$ and $\mathrm{Cov}(X,X) = \mathrm{Var}\,X$.
(b) $\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$.
(c) $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$.

Proof. (a) is immediate.

(b) $aX + b - E(aX+b) = a(X-\mu_1)$, and similarly for Y.

(c) Expand $(X-\mu_1)(Y-\mu_2)$ and use linearity. ■

Theorem 2.3

(a) $\mathrm{Var}\left(\frac{Y-\mu_2}{\sigma_2} - \rho\frac{X-\mu_1}{\sigma_1}\right) = 1-\rho^2$.
(b) $-1 \le \rho \le 1$.
(c) $\rho = \pm 1$ iff $\frac{Y-\mu_2}{\sigma_2} = \pm\frac{X-\mu_1}{\sigma_1}$ with probability 1.

Proof. (a) Expand: $1 - 2\rho\cdot\rho + \rho^2 = 1-\rho^2$.

(b) A variance is nonnegative.

(c) $\rho = \pm 1$ makes the variance in (a) zero, so the variable is constant (zero, since its mean is zero) with probability 1 (Theorem 1.11). ■

Equivalently,

$$|\mathrm{Cov}(X,Y)| \le \sigma_1\sigma_2,$$

a case of the **Cauchy–Schwarz inequality**, $|E(VW)| \le \sqrt{EV^2\,EW^2}$. The correlation measures *linear* association, free of units.

Example 2.4 (Example 2.1 continued)

$EX = \int_0^1 x(x+\frac{1}{2})\,dx = \frac{7}{12}$, $EX^2 = \int_0^1 x^2(x+\frac{1}{2})\,dx = \frac{5}{12}$, so $\mathrm{Var}\,X = \frac{5}{12} - \frac{49}{144} = \frac{11}{144}$. Also $E(XY) = \int_0^1\int_0^1 xy(x+y)\,dy\,dx = \frac{1}{3}$, so $\mathrm{Cov}(X,Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$ and $\rho = -\frac{1}{11}$.

Joint moments and generating functions. The (r,s) joint moment is $m_{r,s} = E(X^rY^s)$. Expanding $e^{t_1X}e^{t_2Y} = \sum_{r,s}\frac{X^rY^s}{r!s!}t_1^rt_2^s$ suggests the **joint mgf** $M(t_1,t_2) = E(e^{t_1X+t_2Y})$, defined when finite for $|t_i| < h_i$.

Theorem 2.4 (Joint mgf)

If $M(t_1,t_2)$ exists near $(0,0)$:
(a) all joint moments exist, $E(X^rY^s) = \partial^{r+s}M/\partial t_1^r\partial t_2^s\big|_{\mathbf{0}}$, and $M(t_1,t_2) = \sum_{r,s}\frac{E(X^rY^s)}{r!s!}t_1^rt_2^s$;

(b) if two pairs have equal joint mgfs near $\mathbf{0}$, they have the same joint distribution.

Proof. (a) Repeat the proof of Theorem 1.7(a) with $e^{|t_1x|+|t_2y|} \le \sum_{\pm,\pm} e^{\pm t_1x \pm t_2y}$ as the dominating function.

(b) As in one dimension, this needs characteristic functions and is beyond these notes. ■

The marginal mgfs are $M_X(s) = M(s,0)$ and $M_Y(t) = M(0,t)$. The joint cumulant generating function $C(t_1,t_2) = \log M(t_1,t_2) = \sum \frac{c_{r,s}}{r!s!} t_1^r t_2^s$ has joint cumulants $c_{1,0} = EX$, $c_{0,1} = EY$, $c_{2,0} = \operatorname{Var} X$, $c_{1,1} = \operatorname{Cov}(X,Y)$, $c_{0,2} = \operatorname{Var} Y$. This follows by expanding $\log(1+A)$ as in one dimension.

Example 2.5

For $f(x,y) = e^{-y}$ on $0 < x < y$ (Example 2.3),

$$M(t_1,t_2) = \int_0^\infty e^{t_1 x} \int_x^\infty e^{-(1-t_2)y}\, dy\, dx = \frac{1}{(1-t_2)(1-t_1-t_2)}, \qquad t_2 < 1,\ t_1 + t_2 < 1.$$

Then $C = -\log(1-t_2) - \log(1-t_1-t_2)$, and differentiating at $\mathbf{0}$ gives $EX = 1$, $EY = 2$, $\operatorname{Var} X = 1$, $\operatorname{Cov}(X,Y) = 1$, $\operatorname{Var} Y = 2$. So $\rho = 1/\sqrt{2}$. Also $M(s,0) = (1-s)^{-1}$: $X \sim \text{Exp}(1)$.

2.3 Conditional distributions and conditional expectation

Motivation. $E[Y \mid X]$ is the best predictor of Y from X, and conditioning computes moments.

Definition (Conditional pmf/pdf)

For $f_1(x) > 0$, $f_{2|1}(y \mid x) = f(x,y)/f_1(x)$. For each fixed x it is a pmf/pdf in y.

In the continuous case $P(X = x) = 0$, so this is justified as a limit: if f and f_1 are continuous,

$$P(c \le Y \le d \mid x \le X \le x+h) = \frac{\frac{1}{h}\int_x^{x+h}\int_c^d f(s,y)\, dy\, ds}{\frac{1}{h}\int_x^{x+h} f_1(s)\, ds} \xrightarrow{h\downarrow 0} \int_c^d f_{2|1}(y \mid x)\, dy.$$

Theorem 2.5

$P(a \le X \le b, c \le Y \le d) = \int_a^b P(c \le Y \le d \mid X = x) f_1(x)\, dx$ (a sum in the discrete case).

Proof. Substitute $f(x,y) = f_{2|1}(y \mid x) f_1(x)$ into the double integral and integrate in y first. ■

Example 2.6 (Example 2.2 continued)

Given no red ball ($X = 0$), $f_{2|1}(y \mid 0) = f(0,y)/f_1(0) = \binom{3}{y}\binom{4}{3-y}/35$, i.e. $4, 18, 12, 1$ over 35 for $y = 0,1,2,3$. This is $H(3;7,3)$: once the red balls are excluded, we are effectively drawing from the 7 non-red balls.

Example 2.7

For $f(x,y) = e^{-y}$ on $0 < x < y$, $f_{2|1}(y \mid x) = e^{-(y-x)}$ for $y > x$: given $X = x$, $Y - x \sim \text{Exp}(1)$. Hence $P(Y > c \mid X = x) = e^{-(c \vee x - x)}$ and, by Theorem 2.5, $P(X \le 1, Y > 2) = \int_0^1 e^{-(2-x)} e^{-x}\, dx = e^{-2}$.

Definition (Conditional expectation and variance)

$E[g(X,Y) \mid X=x] = \int g(x,y) f_{2|1}(y \mid x)\,dy$; $\mu_{2|1}(x) = E(Y \mid X=x)$; $\operatorname{Var}(Y \mid X=x) = E[(Y-\mu_{2|1}(x))^2 \mid X=x] = E(Y^2 \mid X=x) - \mu_{2|1}(x)^2$. Replacing x by X gives the random variables $E(Y \mid X)$ and $\operatorname{Var}(Y \mid X)$.

Conditional expectation is linear and monotone, and functions of the conditioning variable factor out: $E[c(X)g(Y) \mid X=x] = c(x)E[g(Y) \mid X=x]$.

Theorem 2.6

(a) (Tower property) $E[E(Y \mid X)] = E(Y)$.
(b) $\operatorname{Cov}(Y - E(Y \mid X), v(X)) = 0$ for every function v with $Ev(X)^2 < \infty$.

Proof. (a) $\int \left(\int y f_{2|1}(y \mid x)\,dy \right) f_1(x)\,dx = \iint y f(x,y)\,dy\,dx = EY$.

(b) Let $Z = Y - E(Y \mid X)$. Then $E(Z \mid X) = 0$, so $EZ = 0$ by (a), and $E[Zv(X)] = E[v(X)E(Z \mid X)] = 0$. ■

Theorem 2.7 (Best predictor)

Among all functions u, $E[(Y-u(X))^2]$ is minimized by $u(X) = E(Y \mid X)$, and the minimum is $E[\operatorname{Var}(Y \mid X)]$.

Proof. Write $Y - u(X) = (Y - E(Y \mid X)) + v(X)$ with $v = E(Y \mid X) - u(X)$. The cross term has expectation 0 by Theorem 2.6(b), so $E(Y-u)^2 = E(Y - E(Y \mid X))^2 + Ev(X)^2$, minimized when $v = 0$. And $E(Y - E(Y \mid X))^2 = E[E((Y - E(Y \mid X))^2 \mid X)] = E \operatorname{Var}(Y \mid X)$. ■

$E(Y \mid X)$ is called the **regression function** of Y on X.

Theorem 2.8 (Variance decomposition)

$\operatorname{Var} Y = E[\operatorname{Var}(Y \mid X)] + \operatorname{Var}[E(Y \mid X)]$.

Proof. $Y - \mu = (Y - E(Y \mid X)) + (E(Y \mid X) - \mu)$; the two parts are uncorrelated by Theorem 2.6(b). Square and take expectations, using $E[E(Y \mid X)] = \mu$ and Theorem 2.7. ■

Geometrically $Y - EY$ splits into orthogonal pieces: the part explained by X and the residual. This is the basis of regression and ANOVA (Chapter 10).

Example 2.8 (Example 2.1 continued)

$f_{2|1}(y \mid x) = \frac{x+y}{x+1/2}$ on $[0,1]$, so $E(Y \mid X=x) = \frac{x/2+1/3}{x+1/2} = \frac{3x+2}{6x+3}$. The regression is not linear, and it decreases in x, consistent with $\rho < 0$.

Example 2.9 (Checking the decomposition)

In Example 2.7, $E(Y \mid X) = X+1$ and $\operatorname{Var}(Y \mid X) = 1$. So $E[E(Y \mid X)] = EX+1 = 2 = EY$, and $E \operatorname{Var}(Y \mid X) + \operatorname{Var} E(Y \mid X) = 1 + \operatorname{Var} X = 2 = \operatorname{Var} Y$, in agreement with Example 2.5. Also $\operatorname{Cov}(Y - E(Y \mid X), X) = \operatorname{Cov}(Y, X) - \operatorname{Var} X = 0$, as Theorem 2.6(b) requires.

2.4 Independence

Motivation. Random samples are independent observations, and independence is what makes the joint density factor and the variance of a sum equal the sum of variances.

Definition (Independent random variables)

X and Y are independent if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all (Borel) sets A, B.

Theorem 2.9

Each of the following is equivalent to independence:
(a) $F(x,y) = F_1(x)F_2(y)$ for all x, y;
(b) $f(x,y) = f_1(x)f_2(y)$ for all x, y;
(c) $M(t_1,t_2) = M_X(t_1)M_Y(t_2)$ near 0 (when the mgf exists). Moreover (b) holds iff $f(x,y) = g_1(x)g_2(y)$ for *some* functions g_1, g_2.

Proof. Rectangles determine the distribution, so (a)$\Leftrightarrow$ independence.

(a) and (b) are equivalent by differentiation and integration.

(c) follows from the uniqueness of the mgf. If $f = g_1g_2$, integrating gives $f_1 = g_1 \int g_2$, $f_2 = g_2 \int g_1$ and $1 = \int g_1 \int g_2$, so $f = f_1f_2$. ■

The factorization must hold on the whole plane, *support included*: $f(x,y) = 8xy$ on $0 \le x \le y \le 1$ looks like a product, but the triangular support makes X, Y dependent.

Theorem 2.10

If X, Y are independent then (a) $g_1(X)$ and $g_2(Y)$ are independent;
(b) $E[g_1(X)g_2(Y)] = Eg_1(X)\,Eg_2(Y)$;
(c) $\operatorname{Cov}(X,Y) = 0$;
(d) $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. In general $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y + 2\operatorname{Cov}(X,Y)$.

Proof. (a) $\{g_1(X) \in A\} = \{X \in g_1^{-1}(A)\}$.

(b) Factor the double integral.

(c) is (b) with identities and Theorem 2.2(c).

(d) Expand the square. ■

Uncorrelated does not imply independent: if X is uniform on $\{-1,0,1\}$ and $Y = X^2$, then $\operatorname{Cov}(X,Y) = EX^3 = 0$, yet Y is a function of X.

Example 2.10

(a) $f(x,y) = 6e^{-2x-3y}$ on $x, y > 0$ factors as $2e^{-2x} \cdot 3e^{-3y}$ on a product set, so $X \sim \text{Exp}(\frac{1}{2})$ and $Y \sim \text{Exp}(\frac{1}{3})$ are independent.
(b) $f(x,y) = e^{-y}$ on $0 < x < y$ does not factor ($\rho = 1/\sqrt{2}$).
(c) $f(x,y) = \frac{9}{40}(1+x^2y^2)$ on $[-1,1]^2$ (the constant makes it integrate to 1). By the symmetries $(x,y) \mapsto (-x,y)$ and $(x,-y)$, $EX = EY = E(XY) = 0$, so $\operatorname{Cov}(X,Y) = 0$. But $f_1(x) = \frac{9}{20}(1+\frac{x^2}{3})$ and likewise f_2, so $f(1,1) = \frac{9}{20} \neq f_1(1)f_2(1) = \frac{9}{25}$: X and Y are dependent.

2.5 Random vectors

For $\mathbf{X} = (X_1, \dots, X_k)'$ the joint pmf/pdf, marginals (sum or integrate out the other coordinates), cdf and expectations are defined as above. The **mean vector** is $E\mathbf{X} = (EX_1, \dots, EX_k)'$, and the **covariance matrix** is

$$\operatorname{Var}(\mathbf{X}) = E[(\mathbf{X} - \mu)(\mathbf{X} - \mu)'] = \big(\operatorname{Cov}(X_i, X_j)\big)_{i,j}.$$

More generally $\operatorname{Cov}(\mathbf{X}, \mathbf{Y}) = E[(\mathbf{X} - \mu)(\mathbf{Y} - \eta)']$.

Theorem 2.11

For constant matrices A, C and vectors $\mathbf{b}, \mathbf{d}$:

(a) $E(A\mathbf{X} + \mathbf{b}) = AE\mathbf{X} + \mathbf{b}$;

(b) $\operatorname{Var}(A\mathbf{X} + \mathbf{b}) = A \operatorname{Var}(\mathbf{X})A'$ and $\operatorname{Cov}(A\mathbf{X} + \mathbf{b}, C\mathbf{Y} + \mathbf{d}) = A \operatorname{Cov}(\mathbf{X}, \mathbf{Y})C'$;

(c) $\operatorname{Var}(\mathbf{X} + \mathbf{Y}) = \operatorname{Var}\mathbf{X} + \operatorname{Var}\mathbf{Y} + \operatorname{Cov}(\mathbf{X}, \mathbf{Y}) + \operatorname{Cov}(\mathbf{Y}, \mathbf{X})$;

(d) $\operatorname{Var}(\mathbf{X})$ is symmetric and nonnegative definite.

Proof. (a)–(c) follow from $E(CWD) = CE(W)D$ for a random matrix W, applied to $(\mathbf{X} - \mu)(\mathbf{Y} - \eta)'$.

(d) $\mathbf{a}' \operatorname{Var}(\mathbf{X})\mathbf{a} = \operatorname{Var}(\mathbf{a}'\mathbf{X}) \geq 0$. ■

By the spectral theorem $\operatorname{Var}\mathbf{X} = P \operatorname{diag}(\lambda_1, \dots, \lambda_k)P'$ with P orthogonal and $\lambda_1 \geq \cdots \geq \lambda_k \geq 0$. Then $P'\mathbf{X}$ has uncorrelated coordinates with variances λ_i: these are the **principal components**, and the eigenvectors give the main directions of spread.

Generating functions. $M(\mathbf{t}) = E(e^{\mathbf{t}'\mathbf{X}})$ yields all joint moments and determines the distribution. The cgf $C = \log M$ gives $E\mathbf{X} = \nabla C(\mathbf{0})$ and $\operatorname{Var}\mathbf{X} = \nabla^2 C(\mathbf{0})$ (gradient and Hessian).

Conditioning. $f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y} \mid \mathbf{x}) = f_{\mathbf{X},\mathbf{Y}}(\mathbf{x}, \mathbf{y})/f_{\mathbf{X}}(\mathbf{x})$. The tower property, the orthogonality $\operatorname{Cov}(\mathbf{Y} - E(\mathbf{Y} \mid \mathbf{X}), v(\mathbf{X})) = 0$, the matrix decomposition

$$\operatorname{Var}\mathbf{Y} = E[\operatorname{Var}(\mathbf{Y} \mid \mathbf{X})] + \operatorname{Var}[E(\mathbf{Y} \mid \mathbf{X})],$$

and the best-predictor property for $E\|\mathbf{Y} - u(\mathbf{X})\|^2$ carry over with the same proofs.

Independence. Independence iff the joint pdf (or mgf) factors into (any) functions of the separate arguments. Then functions of separate components are independent, product expectations factor, $\operatorname{Var}\sum_i \mathbf{X}_i = \sum_i \operatorname{Var}\mathbf{X}_i$, and $M_{\sum \mathbf{X}_i}(\mathbf{t}) = \prod_i M_{\mathbf{X}_i}(\mathbf{t})$.

Example 2.11

Let X_1, X_2, X_3 be i.i.d. standard exponential and $Y_1 = X_1$, $Y_2 = X_1 + X_2$. Then $\operatorname{Var}(Y_1, Y_2)' = A\,I\,A'$ with $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, i.e. $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, so $\operatorname{Corr}(Y_1, Y_2) = 1/\sqrt{2}$.

2.6 Selected exercises

Motivation. Course exercises for Chapter 2: conditioning, maximal correlation, pairwise vs. mutual independence.

Exercise 2.7 (Pólya's urn)

An urn contains $b \geq 0$ black and $r \geq 0$ red balls, $b + r > 0$. A ball is drawn; set $X_1 = 1$ if it is black and $X_1 = 0$ if red, and return it together with $c \geq 0$ further balls of the same colour. This is repeated, giving $X_2, X_3, \ldots$.
(a) Find the conditional distribution of X_{n+1} given $X_1, \ldots, X_n$.
(b) Find the marginal distribution of X_n, $n \geq 1$.

Solution.

(a) Let $S_n = X_1 + \cdots + X_n$ and $N_n = b + r + nc$. After n draws the urn holds N_n balls, of which $b + cS_n$ are black. The next ball is drawn at random, so

$$P(X_{n+1} = 1 \mid X_1, \ldots, X_n) = \frac{b + cS_n}{N_n} =: M_n, \qquad P(X_{n+1} = 0 \mid X_1, \ldots, X_n) = 1 - M_n.$$

(b)

Step 1. M_n is the proportion of black balls after n draws. If $X_{n+1} = 1$ the urn then holds $M_n N_n + c$ black balls among $N_n + c$; otherwise $M_n N_n$. By (a),

$$E[M_{n+1} \mid X_1, \ldots, X_n] = M_n \frac{M_n N_n + c}{N_n + c} + (1 - M_n) \frac{M_n N_n}{N_n + c} = \frac{M_n (N_n + c)}{N_n + c} = M_n.$$

Step 2. Taking expectations (Theorem 2.6), $EM_{n+1} = EM_n = \cdots = EM_0 = b/(b + r)$.

Step 3. Hence $P(X_{n+1} = 1) = E\big[P(X_{n+1} = 1 \mid X_1, \ldots, X_n)\big] = EM_n = b/(b + r)$. So every X_n is Bernoulli$(b/(b + r))$, as for the first draw, although the X_n are dependent when $c > 0$. ■

Exercise 2.12 (Maximal correlation of linear combinations)

Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be random vectors of dimensions k_1 and k_2 with $\Sigma_{11} = \operatorname{Var} \mathbf{X}_1$ and $\Sigma_{22} = \operatorname{Var} \mathbf{X}_2$ positive definite and $\Sigma_{12} = \operatorname{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \Sigma_{21}'$. Find nonzero $\mathbf{a}_1, \mathbf{a}_2$ maximizing $\operatorname{Corr}(\mathbf{a}_1' \mathbf{X}_1, \mathbf{a}_2' \mathbf{X}_2)$, and the maximum.

Solution.

Step 1. By Theorem 2.11,

$$\operatorname{Corr}(\mathbf{a}_1' \mathbf{X}_1, \mathbf{a}_2' \mathbf{X}_2) = \frac{\mathbf{a}_1' \Sigma_{12} \mathbf{a}_2}{\sqrt{\mathbf{a}_1' \Sigma_{11} \mathbf{a}_1} \sqrt{\mathbf{a}_2' \Sigma_{22} \mathbf{a}_2}}.$$

Put $\mathbf{u} = \Sigma_{11}^{1/2} \mathbf{a}_1$, $\mathbf{v} = \Sigma_{22}^{1/2} \mathbf{a}_2$ (symmetric positive definite square roots) and $K = \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1/2}$. As $\mathbf{a}_1, \mathbf{a}_2$ range over nonzero vectors, so do $\mathbf{u}, \mathbf{v}$, and the correlation equals $\mathbf{u}' K \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$.

Step 2. Let $\lambda_1 \geq 0$ be the largest eigenvalue of the symmetric nonnegative definite matrix $K'K$, with unit eigenvector $\mathbf{e}_1$. By the spectral theorem $\|K\mathbf{v}\|^2 = \mathbf{v}' K' K \mathbf{v} \leq \lambda_1 \|\mathbf{v}\|^2$, and by the Cauchy–Schwarz inequality

$$\mathbf{u}' K \mathbf{v} \leq \|\mathbf{u}\| \, \|K\mathbf{v}\| \leq \sqrt{\lambda_1} \, \|\mathbf{u}\| \, \|\mathbf{v}\|.$$

So the correlation is at most $\sqrt{\lambda_1}$.

Step 3. If $\lambda_1 > 0$, take $\mathbf{v} = \mathbf{e}_1$ and $\mathbf{u} = K\mathbf{e}_1$. Then $\|\mathbf{u}\|^2 = \mathbf{e}_1' K' K \mathbf{e}_1 = \lambda_1$ and $\mathbf{u}' K \mathbf{v} = \lambda_1$, so the correlation equals $\lambda_1 / \sqrt{\lambda_1} = \sqrt{\lambda_1}$. If $\lambda_1 = 0$, then $K = 0$, i.e. $\Sigma_{12} = 0$, and every correlation is 0.

Step 4. Translating back, the maximum is $\sqrt{\lambda_1}$, attained at

$$\mathbf{a}_2 = \Sigma_{22}^{-1/2} \mathbf{e}_1, \qquad \mathbf{a}_1 = \Sigma_{11}^{-1/2} K \mathbf{e}_1 = \Sigma_{11}^{-1} \Sigma_{12} \mathbf{a}_2$$

(and at their positive multiples). Since $K'K = \Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1/2}$ is similar to $\Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$, λ_1 is also the largest eigenvalue of that matrix. The maximum $\sqrt{\lambda_1}$ is the first *canonical correlation*; for $k_1 = k_2 = 1$ it is $|\mathrm{Corr}(X_1, X_2)|$. ■

Exercise 2.15 (Pairwise but not mutually independent)

Let (X, Y, Z) have density

$$f(x, y, z) = \frac{1 - \sin x \sin y \sin z}{8\pi^3}, \qquad 0 \le x, y, z \le 2\pi,$$

and 0 otherwise. Show that any two of X, Y, Z are independent, but X, Y, Z are not mutually independent.

Solution.

Step 1. $f \ge 0$ because $|\sin x \sin y \sin z| \le 1$, and $\int_0^{2\pi} \sin t\, dt = 0$ gives $\iiint f = (2\pi)^3/(8\pi^3) = 1$, so f is a density.

Step 2. Integrating out z and using $\int_0^{2\pi} \sin z\, dz = 0$, the density of (X, Y) is $f_{XY}(x, y) = 2\pi/(8\pi^3) = 1/(4\pi^2)$ on $[0, 2\pi]^2$. Integrating once more, X and Y are uniform on $[0, 2\pi]$. Thus $f_{XY} = f_X f_Y$, and X, Y are independent by Theorem 2.9(b). By symmetry of f in (x, y, z) the same holds for (X, Z) and (Y, Z).

Step 3. Choose $\delta \in (0, \pi/2)$ so small that $\sin t \ge 2^{-1/3}$ for $t \in I = [\pi/2 - \delta, \pi/2 + \delta]$. On the box I^3 we then have $\sin x \sin y \sin z \ge \frac{1}{2}$, so $f \le \frac{1}{2} \cdot \frac{1}{8\pi^3}$ there and

$$P(X \in I, Y \in I, Z \in I) \le \frac{1}{2} \cdot \frac{(2\delta)^3}{8\pi^3} < \Big(\frac{2\delta}{2\pi}\Big)^3 = P(X \in I)P(Y \in I)P(Z \in I).$$

So the product rule fails for the events $\{X \in I\}, \{Y \in I\}, \{Z \in I\}$, and X, Y, Z are not mutually independent. ■